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Dear Editor,

I wish to resubmit the manuscript *Plants in limbo – the theory of detrital photosynthesis* for consideration as a *Theories* article in *Quantitative Plant Biology*. Based on the second round of constructive criticism from reviewer 1, I have rewritten the introduction and provided much additional evidence for detrital physiology beyond detrital photosynthesis, rewritten parts of the discussion, and modified all figures, augmenting each with my own illustrations to guide the reader. I believe I have now fully addressed the reviewer's semantic concerns and demands for more evidence. The few points I rebutted relate to the fact that in this article I "question existing dogma thanks to quantitative analysis" (*Quantitative Plant Biology* author guidelines for *Theories*) and the reviewer seems to prefer existing dogma, as well as the reviewer's seeming unfamiliarity with theoretical approaches to science. The reviewer's requests inevitably led to a much more extensive bibliography that has breached the journal's reference limit. I hope this will be allowed given the extensive review/theory nature of this article. Data and code remain available in the updated repository at github.com/lukaseamus/plant-limbo. I have addressed the reviewers' comments in detail in italics below.

Reviewer: 1

Comments to the Author

Second Round Peer Review for "Plants in limbo – the theory of detrital photosynthesis"

I believe that the paper has improved a lot since its first submission, and most of my comments focus on enhancing the clarity and readability of the study. However, I have some remaining concerns that need to be addressed.

Main concerns remaining:

- Author needs to change the wording throughout from "physiology" to "photosynthesis"- photosynthesis is what the study has measured, not physiology. (i.e., "terrestrial decomposition is probably unaffected by physiology" L531 and in many other places)

I replaced "physiology" with "photosynthesis" where sensible (ll. 33, 86, 572) or removed it altogether. I would like to stress that in terms of word balance I only use the term "physiology" 11 times in the entire manuscript while I use "photosynthesis" 140 times. I also never claim to have measured several aspects of physiology. Since photosynthesis is what powers plant physiology and there are conceivable causal pathways linking plant physiology to decomposition (Figure 4), I maintain that in the remaining 11 cases "physiology" cannot be replaced with "photosynthesis" because I am referring to all aspects of plant physiology. I have expanded on the connection between

detrital photosynthesis and other aspects of plant physiology in a new, second paragraph of the introduction (ll. 51–77).

o From response to reviewer doc: “I use the term “physiology” on purpose because it also encompasses induced defence and wound repair or regrowth which are driven by energy from detrital photosynthesis but may happen in the absence of photosynthesis.”

o ^In response to the above, I do not think that you can assume that just because a plant is photosynthesizing that it is also functioning normally physiologically in all of these ways, or inducing defence. That would need to be measured separately, and has not been to my knowledge (or the author needs to add studies which state that detrital matter can “induce defence” upon detachment). All you’re measuring/accounting for here is photosynthesis. Physiology encompasses all processes and functioning in a living system including osmotic balance, nutrient provisioning, reproduction etc. we do not know enough to link photosynthesis to all of these processes.

I have restructured the introduction to dedicate an entire paragraph to evidence for the wider aspects of physiology (growth, repair, chemical defence, reproduction) in detached macroalgae (ll. 51–77). I conclude the paragraph by stressing the point I made in a previous version of the introduction: “Detrital photosynthesis ... provides the strongest evidence in support of the viability of detritus”, meaning it is the best measure to infer intact plant physiology.

o Changing from physiology to photosynthesis enhances clarity and specificity, and is an easy fix.

- The author contests that macroalgal photosynthesis enables macroalgae to “induce defence mechanisms” during degradation- a link that we have no evidence for. See my comments below. Please remove these highly speculative sections on defence mechanisms- they are not supported by the literature nor by the data presented here.

Since photosynthesis underpins plant physiology but, as you say, it remains unclear what aspect(s) of physiology actually influence(s) decomposition (compare growth and defence causal paths in Figure 4), I believe it is less assuming to use the all-encompassing term “physiology” in the absence of further evidence. Photosynthesis simply cannot directly influence decomposition, so replacing “physiology” with “photosynthesis” in such statements would be wrong on all accounts. I have shown in Wright et al. (2024) that even in complete darkness kelp detritus can remain viable and influence its decomposition in the absence of photosynthesis. When it remains uncertain how the causal path goes, I believe the requested specificity can do more damage.

Line by line Comments:

Introduction (reads a lot more clearly now)

- L35 “Often, floating kelp detritus sinks before it stops photosynthesising”

Reworded in l. 35.

- L38: Sentence starting on this line doesn’t make sense- please re-word

Reworded the first paragraph.

- L42: What does “absent or impaired physiology” mean? Have these alga been experimentally killed?

Reworded to be unequivocal (ll. 39ff.)

- L44: Where is the evidence that photosynthesis is related to induced chemical defence?? I am unaware of any studies showing this.

I added a second paragraph to the introduction (ll. 51–77) which contains a plethora of additional references on other aspects of detrital physiology. Plant physiology is ultimately always underpinned by photosynthesis. There need not be a direct correlation between photosynthetic rate and chemical defence for this to be true. We know based on causal logic that the energy acquired via photosynthesis at one point in time may be allocated to chemical defence at a later stage.

- Paragraph starting L56 is much improved

Thank you. This is now paragraph three of the introduction.

- L95: photosynthetic emancipation of seagrass leaves from the main plant? I find this sentence confusing.

Changed “emancipation” to “independence” (l. 118).

- L118: is this a ref to the dataset or to your other paper?

Wright (2025), now in l. 145, is the detrital photosynthesis dataset. This and Wright (2026a) are Zenodo repositories that are clearly labelled with “Dataset” in the bibliography. Zenodo generates a DOI for each repository which allows them to be cited like papers. Wright (2026b) is the mathematical model that derives from the present manuscript.

- L119: considering macroalga as plants; Bolton 2016 (I think you can provide this reference just the first time you refer to plants as encompassing macroalgae)

I removed the citation from l. 146 but retain it in l. 205 because this is in a new section (Methods) and there clarifies the breadth of the literature search. A reader who happens to skip parts of the introduction may otherwise be confused about my definition of “plant” in the literature search.

- L122: Again, I still do not think that “physiology” is the appropriate term to be used here. Please replace with photosynthesis.

Please see my previous comments and changes.

Methods

- L139 “However, this cannot be considered part of the underlying photosynthetic decay process of interest” what do you mean by this?

Changed to “part of the photosynthetic decay process to be modelled” (l. 165). I am saying that the observed increases in photosynthesis due to confounders, such as increasing light or nutrients, cannot be considered part of the decay process I am trying to model, so I am not going to account for these observations in the design of my model.

- L156: fixing the baseline/intercept at 1 is assuming that baseline photosynthesis before degradation is maximum photosynthetic capacity?

No, it assumes that under unchanged conditions detached photosynthesis cannot exceed attached photosynthesis (l. 162). It makes no assumptions about whether photosynthesis is measured under optimal or sub-optimal conditions. Photosynthetic decay is just about photosynthesis relative to initial values under theoretically unchanged conditions, whatever those conditions may be.

- L166: what do you mean by etc.

Changed to "(e.g. gross photosynthesis)" (l. 192). This means that gross photosynthesis is just one of many examples where the photosynthetic response variable is bounded by zero. Others include F_v/F_m , chlorophyll etc.

- L243: photosynthesis encompasses the list of response variables L185, but not chlorophyll? Can you clarify here?

I am not sure what to clarify here. All photosynthesis and chlorophyll variables are listed separately (ll. 212–215) and it is explained why this grouping is included in the model (ll. 269–272).

- L248: but water was not included in the model?

No, for reasons laid out (l. 272), the presence of water was not included.

Results

- L320: what do you mean by "on a mass basis" vs. on a leaf basis? Shouldn't you just standardize to mass for clarity/simplicity?

I report both estimates because leaves of *Halophila ovalis* and leaf clusters of *Amphibolis antarctica* are quite uniform and leaf-based rates may be less variable than mass-based rates (see data at https://github.com/lukaseamus/plant-limbo/blob/main/Seagrass/RDS/P_summary.rds). Leaf-based photosynthesis is sometimes reported in the literature and may be useful for some readers who want to estimate photosynthesis based on leaf count alone. It is also more easily converted to area-based photosynthesis due to fairly consistent leaf area in these seagrasses. Finally, mass- and leaf-based photosynthesis can give readers complementary insights into different levels of photosynthetic decay: when leaf-based photosynthesis declines this can be due to a decline in leaf mass as well as photosynthesis per leaf mass. I added some of this rationale to the supplementary methods after Equation S1, starting with "I also calculated leaf-based P_{max} ".

- L330: move to discussion.

My observation of periphyton is a result which modifies the predictions I make for seagrasses as a whole (ll. 359f.). I have added species names in parentheses (l. 360) to show that my predictions for seagrasses are based on my own results, not interpretation of the literature. The initial group-level prediction for seagrasses links the empirical results section nicely to the meta-analysis where this prediction is formalised. At no stage do I cite the literature or make predictions that are not based on the data I present, so I do not believe this is appropriate for the discussion section.

- L365: move to discussion.

This is also a plain result. My meta-analysis clearly shows that photosynthetic half-life is typically 30–50% longer when based on chlorophyll than when based on photosynthesis (see the parameter gamma in Table 2), so the statement “Measurements based on chlorophyll therefore tend to overestimate the longevity of detrital photosynthesis” is not a subjective interpretation.

- L370: remind me what do you mean by “new plants”? Add this context to this sentence, unclear where you’ve discussed what is meant by new plants previously

I had previously mentioned this in the methods but elaborated with some examples (ll. 301f.). I also reworded to “unobserved plant groups” (ll. 384, 401) to be less ambiguous. See ll. 409, 464, 467 for further context on predictions for new occurrences.

Discussion

- L457 “over measurable”? Is this a typo?

Not a typo. Added commas to clarify sentence structure (ll. 489f.). The sentence reads “ranges from x, over y, to z”.

- L458: I think that you should re-word this sentence. Macroalgal decomposition is affected by detrital photosynthesis, not determined.

I already state that macroalgal decomposition is substantially affected by detrital photosynthesis in the previous sentence (l. 490). Then I make a subjective speculation that decomposition may even be determined by detrital photosynthesis, hence the use of “it is possible”. I replaced “possible” with “conceivable” (l. 491) to clarify the subjective nature of this speculation.

- L500: I think this is an awkward sentence that could be re-worded for clarity

Reworded (ll. 537f.).

- L504: This sentence also does not make sense to me and should be explained more

As in the introduction, I replaced “emancipation” with “independence” (l. 541). I expect the reader to make the link to the relevant section in paragraph four of the introduction (ll. 118ff.).

- L511: Greater than what... to aid in clarity extend this thought

Added “...than predicted by my meta-analysis” (l. 548). The point I am making is that my formal analysis is only as good as the data and the data are skewed towards measurements of terrestrial plant leaves in water.

- L514: so to check, hydrogen peroxide only accumulates in freshwater but not marine plants? Can you clarify?

There is no evidence for seagrasses. I added "...which may be absent in seagrass leaves." (ll. 552f.) to clarify this.

- L533: "Since the majority of vascular plant decomposition occurs in the absence of physiology, its main intrinsic predictor must be constitutive defence while that of macroalgal decomposition is likely induced defence (Figure 4)."

- How can any biological process occur in the absence of physiology? I do not understand the reasoning behind this sentence and it needs to be modified (see comments related to figure 4 below)

Changed to "photosynthesis" (l. 572) as requested by you above. I never once mention detritivore physiology, so the 11 mentions of the term "physiology" clearly refer to plant physiology.

- Why do you say that defences are the main predictor of degradation?

I have re-expressed "induced defence" as "physiologically induced resistance to decay" (l. 574) or "physiologically dependent recalcitrance" for short (Figure 4), to allow for other mechanisms such as detrital growth. I believe this has now been appropriately expanded on in the second paragraph of the introduction (ll. 51–77) and the updated section of this paragraph (ll. 571–587).

- L536: IMPORTANT It is much more likely that the increase in phlorotannin content in kelp detritus (assuming you mean as degradation progresses) is a by-product of degradation wherein the microbes degrading the detritus preferentially consume the more labile compounds in macroalgal detritus first, resulting in an increase in refractory compounds in the remaining detritus. This is a much more plausible explanation, and there is ample microbial ecology literature which supports this mechanism (microbes go for the most labile compounds first during decomposition of organic matter), but no literature on the proposed mechanism here which is presumably that the macroalgae begin producing more refractory compounds (phlorotannin's etc) after detachment as an "induced defence" through some unknown mechanism.

I am aware of the microbial ecology literature you refer to, mostly in relation to the preferential consumption of labile, nitrogen-rich organic matter, and agree that this is a possible interpretation for detritus sensu stricto. On a similar note, de Bettignies et al. (2020) suggested that the increase in phlorotannins can simply be attributed to easier extraction from degraded kelp cells. But in Wright et al. (2022) I have already speculated on the possibility of chemical defence in kelp detritus. I am just reinforcing these speculations here. Quite to the contrary of being "a much more plausible explanation", I believe treating macroalgal detritus as dead organic matter sensu Moore et al. (2004) is not defensible in light of the evidence I present. I also disagree that the induced defence mechanism is unknown. There is a plethora of data on induced phlorotannin production in excised seaweed tissues. I now elaborate on this evidence in the second paragraph of the introduction where I cite many additional studies (ll. 56–66). If, after reading the current manuscript, you still believe that microbes interacting with passive detritus sensu stricto is a better model of macroalgal decomposition than the one I put forward, I am not sure I can make a stronger case to "question existing dogma thanks to quantitative analysis" (Quantitative Plant Biology author guidelines) than I already have.

- "other induced defences such as hydrogen peroxide release have yet to be measured in detritus"

- ♣ Getting into these speculative sections on induced defence really muddies the water and is beyond the scope of this paper- please remove. You need some basal evidence to propose something like this

at least, and we currently do not have any evidence (while we have plenty of evidence for the much more plausible mechanism, related to microbial transformation of detritus, stated above)

I have added the requested “basal evidence” on hydrogen peroxide defence in paragraph two of the introduction (ll. 56–60). I have also reworded this discussion paragraph (ll. 574–583). I believe the literature and my quantitative analysis of detrital photosynthesis are enough to go on to propose a new conceptual model. Of course, it is one of many interpretations and speculative until corroborated by future observations. So is the nature of theory.

- L541: “In this case...” confusing sentence, can you re-word it for clarity? Or maybe just remove?

Reworded (ll. 586f.).

- L548: must should

Replaced “must” with “should” (l. 593).

- L550: in what experiment? What do you mean by this?

Reworded (ll. 594f.).

- L558: what is the evidence for this? “the majority of tissue that is abscised or eroded likely exhibits impaired photosynthesis from the start” if there is no evidence then please remove

I have added evidence (ll. 604f.).

- L559: “Ignorance of the causation introduced by the presented theory (Figure 4) would lead to the erroneous conclusion that decomposition rates measured on excised tissue are representative”
 - Representative of what? In what sense? Please clarify. Where do you show that excised portions of macroalgal tissues degrade/photosynthesize differently than whole plants?

Added specificity (ll. 602ff.).

- L564: in what way would these variables be controlled for? Shouldn’t we just aim to explain or account for variables such as temp, light in models?

Elaborated (ll. 605–624). I believe it is more important to understand the causal mechanism behind effects than including variables in models and measuring their effects.

- L577: “can hence be added as a predictor variables to models predicting decomposition” ?
- L578: along with other biotic and abiotic variables

Reworded (ll. 630f.).

- L581: again, why is induced defence being brought up here? It is not in the scope of this article.

This section lays out priorities for future research. I am sure you agree that chemical defence in detritus is a wide knowledge gap that is directly relevant to the connection between detrital photosynthesis and decomposition.

- L582: how would 3 help accomplish better understanding when detrital photosynthesis would always be a factor in situ? Maybe better suggest experiments in complete darkness to accomplish this aim?

Detrital photosynthesis may not always be relevant in nature. As elaborated in the penultimate paragraph of the discussion, much of the detritus produced by erosion is probably photosynthetically impaired or even dead (ll. 602–605, but again here we need more data as suggested in point 3). In this case, a decomposition experiment that pre-kills the detritus or uses senescent material that is about to be eroded/abscised may be more realistic. Furthermore, the direct effect of plant physiology on decomposition has not been quantified except in the few studies I cite. It is important to be able to put a number on this effect rather than just say it exists theoretically as I have done here. Experiments in darkness are also important but, as I have shown in Wright et al. (2024) and others have shown elsewhere, fresh kelp can remain alive in darkness for considerable periods, so there is a legacy effect of photosynthesis.

- Sorry if I missed it but did you discuss the implications of the lack of data for certain groups on your analysis? From L247 methods? That we need data on dark/light for certain groups to better understand trends.

Added (ll. 640ff.).

Figures

- L402: Can you label the plot as panel a and b, and refer to a and b separately in the caption? I think this would help with understanding and is standard practice for most journals.

I understand the need for separate panels when different data are displayed and the connection is ambiguous, or when multiple similar plots are combined (e.g. two scatterplots). However, here everything is part of one figure arranged into four colour-coded columns, corresponding to the plant groups. The distributions are well-labelled and correspond to the scatterplots above. I am inclined not to visually break up this alignment by splitting it into vertical panels. The caption describes “points”, “lines and intervals”, and “distributions”, all of which are very distinct and common visual elements.

- L467: It is best practice to define all acronyms/symbols in each table/figure caption so that a reader can interpret each figure/table on its own- please define p, k, etc. in the caption (also goes for Table 1 and other figures- this helps with easy interpretation, I know you’ve defined these variables elsewhere, best practice to define again in each figure/table caption)

I have repeated definitions as requested (ll. 372f. 461, 500ff.).

- Comments for figure 4
 - This figure is quite busy- what do the arrows coming out of the seaweeds and plants represent? I’d suggest removing them. Overall, I would try to simplify this figure.

I added a sentence to clarify the meaning of the arrows (ll. 507f.). Without the arrows, the illustration is not clear because the temporal sequence becomes obscure. I have changed the illustrations in order to make the figure clearer. I also explain in the caption that the boxes are insets (l. 505).

o Not clear why you say physiology induces defence in seaweeds? Are there studies that show that as seaweeds decay they induce defences against herbivory/microbes? Or does photosynthesis in itself confers a defence against degradation? Even so, can we assume that this is always the case? What are these induced defence mechanisms anyways? It seems as though that the persistence of biomass related detrital photosynthesis has more to do with the tissues just functioning normally/semi-normally allowing some growth to offset degradation, but we do not have reason to believe that photosynthesis itself is directly protecting against degradation through induced defence mechanisms.

Please see my answers to all your previous related comments. I have added a sentence to the caption, further emphasising that this is a theoretical figure and more empirical work is needed (ll. 508f.). I use adjectives such as "conceptual", "schematic" and "speculative" throughout the caption to describe the nature of this figure. Please keep in mind that this paper will be published as a "Theories" article.

o Abiotic what? Drivers of decay? Overall I think this figure helps with understanding however the boxes (conventional model vs model w detrital photosynthesis) do not make sense to me. Why is the arrow going from abiotic to decomposers?

The caption explains directed acyclic graphs (DAGs) and their purpose in causal modelling (ll. 505f.). Nodes in DAGs are variables, so "Abiotic" refers to abiotic variables, such as light, temperature, oxygen etc. There is a causal link between abiotic variables and decomposers. Just think of how temperature affects saprotroph metabolism or UV light kills bacteria.

o What do you mean by constitutive defence in vascular plants?

Constitutive defences can be structural or chemical, but the defining feature is that they are always active, whereas induced defences are only activated when needed. Induced defences are always dependent on physiology because the plant has to respond to the attack. Some constitutive defences such as continual hydrogen peroxide release are dependent on physiology, whereas constitutive structural defences such as lignin are not. For tannins the distinction becomes even more blurry since they need to be maintained by physiology, even when used as a constitutive defence, but they may also persist for a considerable time in dead leaves. I realise that due to this ambiguity, "constitutive" and "induced" don't quite capture the broader meaning I want to convey, so I have reworded "constitutive defence" and "induced defence" as "physiologically independent recalcitrance" and "physiologically dependent recalcitrance" respectively and define "recalcitrance" as "resistance to decomposition" (l. 504).

Reviewer: 2

Comments to the Author

The author has addressed all of my comments and suggestions satisfactorily. I have no further comments.

Thank you.

Sincerely,
Luka Seamus Wright